Galaba Vamsi

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Objective

Enthusiastic Mechatronics undergraduate at IIT Bhilai with strong technical, leadership, and project management skills. Eager to contribute to software development, robotics, and engineering projects, applying a hands-on approach and a passion for making technology accessible and impactful.

Education

IIT Bhilai
Bachelor of Technology in Mechatronics *2023 – 2027*
Relevant Coursework: Data Structures and Algorithms, Machine Learning, Robotics Systems, C++, Python Programming, ROS2, Linear Algebra, Probability and Statistics.

Air Force School Hebbal
CBSE Class 12
Scored 96%, with a strong focus on Mathematics and Computer Science.

Air Force School Yelahanka
CBSE Class 10
Scored 94.4% with distinction in STEM subjects.

Technical Skills

Programming Languages: Python, C, C++, JavaScript, Dart

Web Development: React.js, Node.js, Three.js, Tailwind CSS, Flask, Bootstrap

App Development: Flutter

Robotics: ROS2, NVIDIA Isaac, Drone Development, Navigation Systems, MuJoCo

Machine Learning: Machine Learning, Deep Learning, Computer Vision, TensorFlow, Data Structures & Algorithms

Data Analysis: Numpy, Pandas

CAD Design: SolidWorks, AutoCAD

Simulation & Modeling: MuJoCo, NVIDIA Isaac Sim, Unreal Engine

Other Tools: UI/UX (Figma, Adobe XD), Blender

Projects

IIT Bhilai
RIS Simulator *2025 – Present*
Developed a GUI-based Reconfigurable Intelligent Surface (RIS) simulator for advanced testing and coverage analysis, integrating Python, Go, and Qt/C++. Contributed to RIS fabrication support and performance evaluation.

IIT Bhilai
Shared Action Autonomous Smart Wheelchair *2025 – Present*
Led development of a smart wheelchair integrating EEG/EMG, joystick, and voice controls for enhanced accessibility. Applied deep learning techniques for signal processing and control integration. Utilized NVIDIA Isaac Sim and MuJoCo for simulation and testing.

IIT Bhilai
Polaris Website *2024*
Designed and deployed IIT Bhilai's official freshers' orientation website, featuring interactive dashboards, event updates, and resource hubs, improving onboarding experience.

Serenity Mental Health App *Ongoing*
Flutter-based app offering stress analysis, interactive games, journaling, fitness tracking, and meditation tools, aiming to enhance student mental health and well-being.

IIT Bhilai - Epsilon Club
Mars Rover Project *2024*
Contributed to CAD modeling and navigation systems development for Epsilon Club's Mars Rover project. Integrated ROS2 with NVIDIA Isaac Sim and MuJoCo for simulation and real-time testing.

Experience

IIT Bhilai
Research Intern *May 2025 – Present*
Working on RIS technologies, developing GUI simulators and supporting hardware fabrication, contributing to research publications and lab automation.

IIT Bhilai
Secretary, Film and Photography Society *Apr 2024 – Apr 2025*
Elected as FPS Secretary; led initiatives that improved the club's Inter-IIT ranking from 17th to 6th place. Organized workshops, competitions, and inter-IIT participation, significantly enhancing club visibility and performance.

IIT Bhilai
Student Mentor *May 2024 – Apr 2025*
Guided first-year students academically and socially, promoting a supportive environment. Helped with onboarding and academic resources, ensuring smooth transition to college life.

Swiggy Campus CEO *Apr 2024 – Sep 2024*
Executed marketing strategies, data-driven consumer engagement, and promotional campaigns to enhance brand presence and adoption on campus.

IIT Bhilai
Polaris Website Developer *May 2024 – Jul 2024*
Developed and maintained the official orientation website, collaborating with administrative and student teams to ensure real-time information and user-friendly design.

Achievements

FPS Inter-IIT Ranking: Improved IIT Bhilai's FPS ranking from 17th to 6th place in a single year as Secretary, demonstrating leadership and strategic planning.

Toyota Hackathon: Top 10 placement for Automated Car Safety System prototype.

ATL Hackathon: 2nd Prize for BMTC Ticketing App addressing COVID safety and efficient transport.

Epsilon Club: Contributed to Mars Rover Project's CAD modeling and system integration.

Graphic Design: Shortlisted for regional round of National Skill India Competition.

Badminton: 3rd Prize in Under-19 tournament by Decathlon and Indian Air Force.

Extracurricular Activities

Robotics: Active participant in Epsilon Club, contributing to Mars Rover and warehouse drone projects.

Film & Photography: Produced videos, led club events, and managed social media for FPS.

Reading: Avid reader of science fiction and popular science books, including The Martian, Three-Body Problem series, and Dune.

Gaming & Creative Arts: Skilled in Blender and Unreal Engine for 3D modeling and simulation; enjoys competitive card games and game design.